



2014 Half-Yearly Examination

FORM VI

MATHEMATICS 2 UNIT

Wednesday 26th February 2014

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 85 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 75 Marks

- Questions 11–15 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 5 per boy
- Multiple choice answer sheet
- Candidature — 90 boys

Examiner
GMC

SECTION I - Multiple Choice

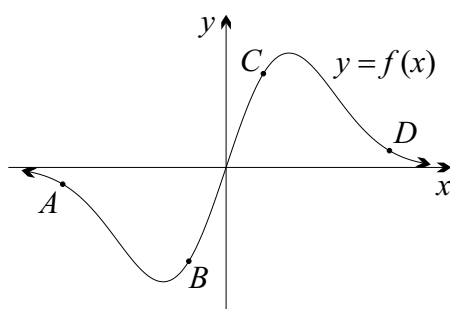
Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Given that $y = \frac{1}{x}$, which of the following statements is true?

- (A) $\frac{dy}{dx} = \frac{1}{x^2}$
- (B) $\frac{dy}{dx} = -\frac{1}{x^2}$
- (C) $\frac{dy}{dx} = \frac{2}{x^2}$
- (D) $\frac{dy}{dx} = -\frac{2}{x^2}$

QUESTION TWO



The graph of $y = f(x)$ is shown above.

Which of the labelled points satisfies $f(x) < 0$ and $f''(x) > 0$?

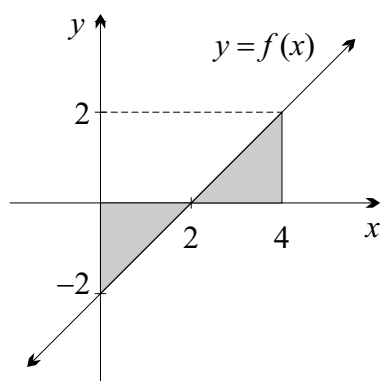
- (A) A
- (B) B
- (C) C
- (D) D

QUESTION THREE

The value of the limit $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$ is given by:

- (A) 0
- (B) $\frac{1}{3}$
- (C) $\frac{1}{6}$
- (D) None of the above.

QUESTION FOUR



A linear function $y = f(x)$ is graphed above.

Which of the following expressions represents the area of the shaded region?

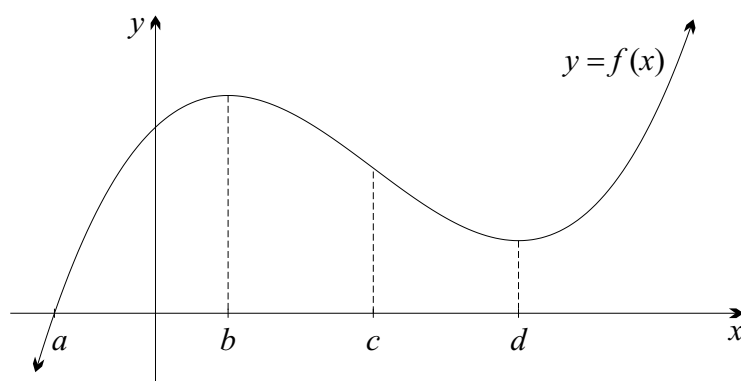
- (A) $\int_0^4 f(x) dx$
- (B) $-\int_0^4 f(x) dx$
- (C) $2 \int_0^2 f(x) dx$
- (D) $2 \int_2^4 f(x) dx$

QUESTION FIVE

What are the coordinates of the focus of the parabola $x^2 = -4y$?

- (A) $(0, 1)$
- (B) $(1, 0)$
- (C) $(0, -1)$
- (D) $(-1, 0)$

QUESTION SIX



The diagram above is a graph of $y = f(x)$.

For what values of x is the function $y = f'(x)$ positive?

- (A) $x > 0$
- (B) $x < b$ or $x > d$
- (C) $x > a$
- (D) $b < x < d$

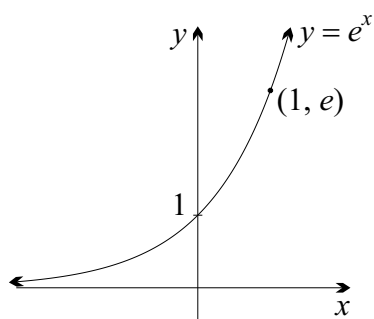
QUESTION SEVEN

What is the value of the definite integral $\int_0^1 e^{-x} dx$?

- (A) $\frac{e-1}{e}$
- (B) $\frac{1-e}{e}$
- (C) $\frac{e+1}{e}$
- (D) $\frac{1}{e}$

QUESTION EIGHT

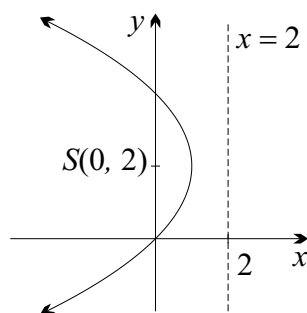
The graph of $f(x) = e^x$ is shown below.



Which one of the following statements is false?

- (A) $f(x) > 0$
- (B) $f'(x) > 0$
- (C) $f(x) = f'(x)$
- (D) $f(-x) = -f(x)$

QUESTION NINE



A parabola with focus $S(0, 2)$ and directrix $x = 2$ is shown above.
Which of the following is the equation of the parabola?

- (A) $(y - 2)^2 = 4(x - 1)$
- (B) $(y - 2)^2 = -4(x - 1)$
- (C) $(y - 2)^2 = 8x$
- (D) $(y - 2)^2 = -8x$

QUESTION TEN

The continuous function $y = f(x)$ has the properties that $\int_a^c f(x) dx = 7$ and $\int_b^c f(x) dx = -4$. Given that $a < b < c$, what is the value of $\int_a^b f(x) dx$?

- (A) 11
- (B) -11
- (C) 3
- (D) -3

_____ End of Section I _____

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.	Marks
(a) Calculate $3e^{-2}$ correct to 2 decimal places.	1
(b) Differentiate the following with respect to x :	
(i) $2x^2 + 5$	1
(ii) $x^{\frac{1}{3}}$	1
(iii) $(4x + 1)^6$	2
(c) Find a primitive for each of the following:	
(i) $3x^5$	1
(ii) e^{-2x}	1
(iii) $\sqrt{x}$	2
(d) Write down the equation of the locus of the point $P(x, y)$ that is:	
(i) 3 units from the point $(-2, 1)$,	1
(ii) 4 units below the line $y = 1$.	1
(e) Sketch a graph of $y = e^{-x} + 2$ clearly showing the asymptote and y -intercept.	2
(f) Consider a curve whose first derivative is given by $y' = 3x + 2$. For what value of x is the curve stationary?	1
(g) Consider a curve whose second derivative is given by $y'' = 2x + 4$. For what values of x is the curve concave up?	1

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**

(a) Simplify $\frac{e^{3x+2}}{e^x}$.

1(b) Differentiate the following with respect to x :

(i) $y = (2x - 1)(3x + 2)$

2

(ii) $y = \frac{x^2 - 2}{x}$

2

(iii) $y = \sqrt{x^2 + 1}$

2

(c) Evaluate $\int_{-1}^2 (4 - x^2) dx$.

2(d) A parabola has equation $x^2 = 16y$.

(i) Write down the coordinates of the vertex.

1

(ii) Find the coordinates of the focus.

1

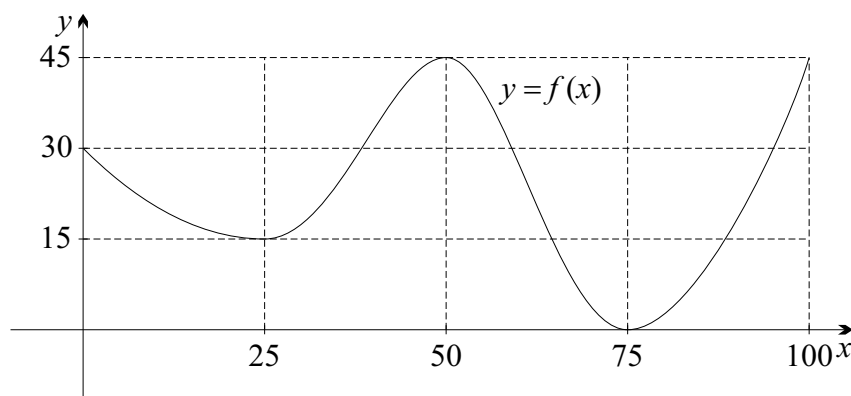
(iii) Find the equation of the directrix.

1

(iv) Sketch the parabola clearly showing the vertex, focus and directrix.

1

(e)

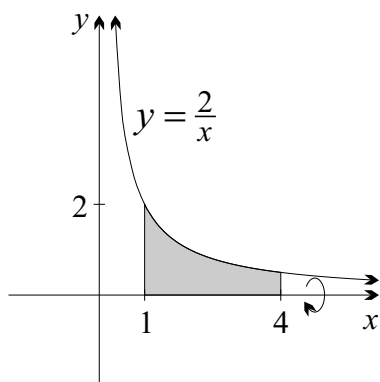
The diagram above shows the graph of $y = f(x)$.

(i) Copy and complete the table below.

1

x	0	25	50	75	100
$f(x)$					

(ii) Hence estimate $\int_0^{100} f(x) dx$ using Simpson's rule with five function values.**1**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Consider the function $y = x^3 - 6x^2 + 9x - 1$.(i) Show that $\frac{dy}{dx} = 3(x-1)(x-3)$ and find $\frac{d^2y}{dx^2}$. 2(ii) Find the coordinates of any stationary points and determine their nature. 2(iii) Find the coordinates of the point of inflexion. You must show that it is a point of inflexion. 2(iv) Sketch the graph of the function, clearly showing all stationary points, the point of inflexion and the y -intercept. Do NOT attempt to find any x -intercepts. 2(b) 2

The region bounded by $y = \frac{2}{x}$, the x -axis and the lines $x = 1$ and $x = 4$ is shown above.

Find the volume of the solid generated when this region is rotated about the x -axis.

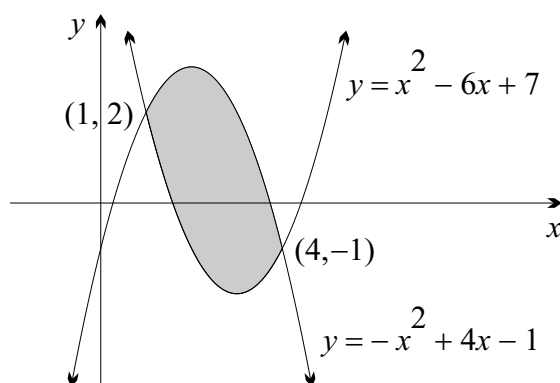
(c) A curve has gradient function $\frac{dy}{dx} = 6x^2 - 3$ and passes through the point $(1, 5)$. 2

Find the equation of the curve.

(d) Find the value of k if $\int_2^k (x-1) dx = 4$ and $k > 2$. 3

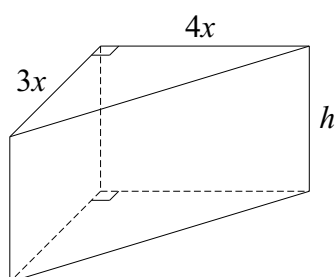
QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Find the equation of the tangent to the curve $y = 1 - e^{-x}$ when $x = 1$. **3**(b) (i) Express the equation $y^2 + 4y - 3x - 5 = 0$ in the form $(y - k)^2 = 4a(x - h)$. **2**(ii) Hence find the coordinates of the focus and the equation of the directrix of the parabola $y^2 + 4y - 3x - 5 = 0$. **2**(c) Using the quotient rule, or otherwise, find the derivative of $y = \frac{2x^2}{e^{3x}}$. Express your answer in simplest form. **2**

(d)

The diagram above shows the curves $y = -x^2 + 4x - 1$ and $y = x^2 - 6x + 7$.(i) By solving a pair of simultaneous equations, show that the points of intersection of the two curves are $(1, 2)$ and $(4, -1)$. **1**(ii) Hence calculate the shaded area between the curves. **2**(e) (i) Differentiate $y = e^{x^2}$. **1**(ii) Hence evaluate $\int_0^1 6xe^{x^2} dx$. **2**

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Consider the exponential function $y = e^{-3x}$.(i) Find y' and y'' .**2**(ii) Show that $y = e^{-3x}$ satisfies the equation $3y = 5y' + 2y''$.**1**(b) The locus of a point $P(x, y)$ is a circle. The distance of P from $A(8, -16)$ is three times the distance of P from the origin.(i) Find the equation of the locus of P .**2**(ii) Hence write down the centre and radius of the circle described by P .**1**

(c)



A closed metal box is in the shape of a prism with a right-angled triangular cross section. The surface area of the box is 240 cm^2 and the perpendicular sides of the triangular cross section are in the ratio $3 : 4$. Let the dimensions of the box be $3x$, $4x$ and h as shown in the diagram above.

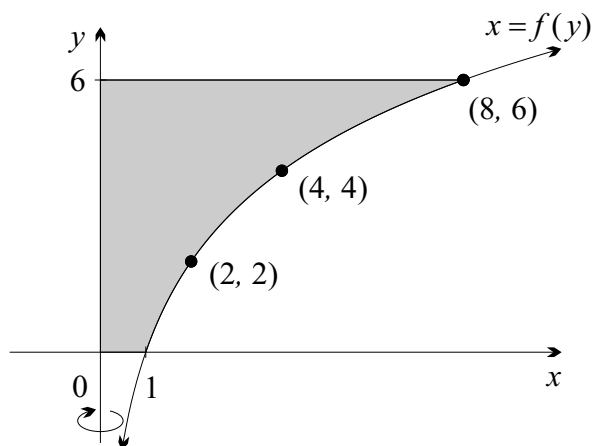
(i) Show that the surface area, S , of the box is given by $S = 12x^2 + 12xh$.**1**(ii) Show that the volume of the box, V , is given by $V = 120x - 6x^3$.**2**

(iii) Hence find the greatest possible volume of the box in exact form.

2**QUESTION CONTINUES ON THE NEXT PAGE**

QUESTION FIFTEEN (Continued)

(d)



The curve $x = f(y)$ passes through the points $(1, 0)$, $(2, 2)$, $(4, 4)$ and $(8, 6)$ as shown in the diagram above.

- (i) Using the trapezoidal rule with four function values, estimate the volume formed by rotating the region bounded by $x = f(y)$, the x -axis, the y -axis and $y = 6$ about the y -axis. **3**
- (ii) Does the trapezoidal rule under-estimate or over-estimate the volume of the solid formed in part (i)? Justify your answer. **1**

_____ End of Section II _____

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

- | | |
|------|-------|
| 1. B | 6. B |
| 2. B | 7. A |
| 3. C | 8. D |
| 4. D | 9. B |
| 5. C | 10. A |

/10

11. a) $3e^{-2} = 0.41$ (2 d.p.)

✓

must show correct to 2 d.p.

b) i) $\frac{d}{dx}(2x^2 + 5) = 4x$

✓

ii) $\frac{d}{dx}(x^{1/3}) = \frac{1}{3}x^{-2/3}$

✓

or $\frac{1}{3x^{2/3}}$

iii) $\frac{d}{dx}((4x+1)^6) = 24(4x+1)^5$

✓✓

one mark for $6(4x+1)^5$

c) i) $\int 3x^5 dx = \frac{x^6}{2} + C$

✓

ii) $\int e^{-2x} dx = -\frac{1}{2}e^{-2x} + C$

✓

iii) $\int x^{1/2} dx = \frac{2}{3}x^{3/2} + C$

✓✓

NB: Constant of Integration not required here.

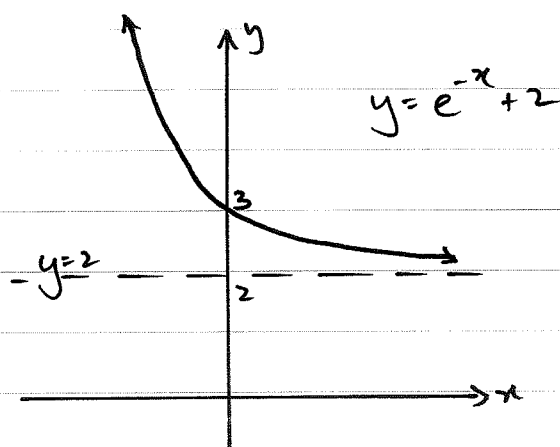
d) i) $(x+2)^2 + (y-1)^2 = 9$

✓

ii) $y = -3$

✓

e)



✓✓

-1 for incorrect shape, asymptote or missing y-intercept. Do not need a separate point.

f) let $y' = 0$

$$3x + 2 = 0$$

$$x = -2/3$$

So the curve is stationary when $x = -2/3$ ✓

g) let $y'' > 0$

$$2x + 4 > 0$$

$$2x > -4$$

$$x > -2$$

So the curve is concave up when $x > -2$ ✓

12. a) $\frac{e^{3x+2}}{e^x} = e^{2x+2}$ ✓

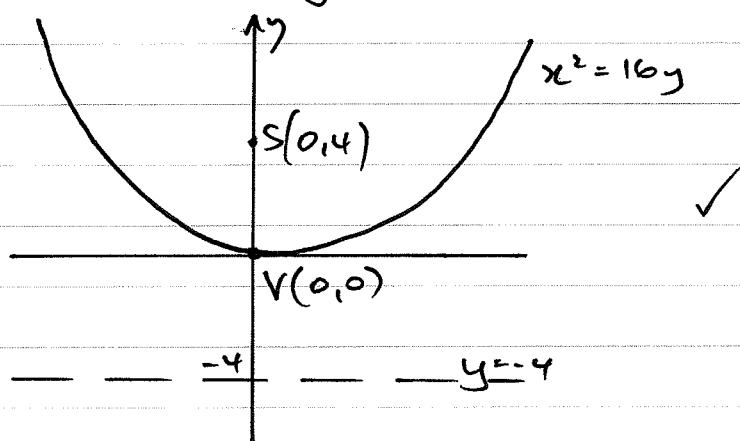
b) i) $\frac{d}{dx}((2x-1)(3x+2))$
 $= \frac{d}{dx}(6x^2 + x - 2)$ ✓
 $= 12x + 1$ ✓

ii) $\frac{d}{dx}\left(\frac{x^2-2}{x}\right)$
 $= \frac{d}{dx}\left(x - \frac{2}{x}\right)$ ✓
 $= 1 + \frac{2}{x^2}$ ✓ or $1 + 2x^{-2}$

iii) $\frac{d}{dx}(\sqrt{x^2+1})$
 $= \frac{d}{dx}((x^2+1)^{1/2})$ ✓
 $= \frac{1}{2}(x^2+1)^{-1/2} \times 2x$ ✓
 $= \frac{x}{\sqrt{x^2+1}}$ or $x(x^2+1)^{-1/2}$ ✓

$$\begin{aligned}
 c) \quad \int_{-1}^2 (4-x^2) dx &= \left[4x - \frac{x^3}{3} \right]_{-1}^2 \quad \checkmark \\
 &= \left(8 - \frac{8}{3} \right) - \left(-4 + \frac{1}{3} \right) \\
 &= 9 \quad \checkmark
 \end{aligned}$$

- d)
- i) Vertex $(0,0)$ $\checkmark$
 - ii) Focus $(0,4)$ $\checkmark$
 - iii) Directrix: $y = -4$ $\checkmark$
 - iv)



e) i)

x	0	25	50	75	100
$f(x)$	30	15	45	0	45

 $\checkmark$

$$\begin{aligned}
 ii) \quad \int_0^{100} f(x) dx &\doteq \frac{50-0}{6} \{30 + 4 \times 15 + 45\} + \frac{100-50}{6} \{45 + 4 \times 0 + 45\} \\
 &\doteq 1875 \quad \checkmark
 \end{aligned}$$

13. a) $y = x^3 - 6x^2 + 9x - 1$

i) $\frac{dy}{dx} = 3x^2 - 12x + 9$
 $= 3(x^2 - 4x + 3)$
 $= 3(x-1)(x-3)$ as reqd. ✓

$\frac{d^2y}{dx^2} = 6x - 12$ ✓

ii) let $\frac{dy}{dx} = 0$

$3(x-1)(x-3) = 0$

$x = 1, x = 3$ } Stat pts at $(1, 3)$ and $(3, -1)$ ✓
 $y = 3, y = -1$ }

when $x = 1$, $\frac{d^2y}{dx^2} = -6$

< 0

so $(1, 3)$ is a max ✓

when $x = 3$, $\frac{d^2y}{dx^2} = 6$

> 0

so $(3, -1)$ is a min ✓

iii) let $\frac{d^2y}{dx^2} = 0$

$6x - 12 = 0$

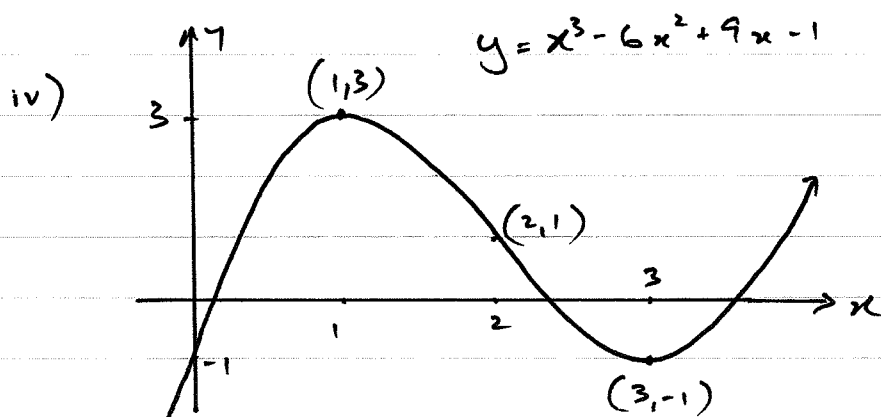
$x = 2, y = 1$

test:

x	1	2	3
$\frac{d^2y}{dx^2}$	-6	0	6

↖ change of sign ↗ ✓

So $(2, 1)$ is a point of inflexion ✓



-1 for any missing point or y-intercept. ✓✓

$$\begin{aligned}
 \text{b) } V &= \pi \int_1^4 y^2 dx \\
 &= \pi \int_1^4 4x^{-2} dx \quad \checkmark \\
 &= -4\pi [x^{-1}]_1^4 \\
 &= -4\pi \left(\frac{1}{4} - 1\right) \\
 &= 3\pi \quad \checkmark
 \end{aligned}$$

So volume is 3π units³

$$\begin{aligned}
 \text{c) } \frac{dy}{dx} &= 6x^2 - 3 \\
 y &= 2x^3 - 3x + C \quad \checkmark \\
 \text{when } x=1, y=5 \\
 5 &= 2 - 3 + C \\
 C &= 6 \\
 y &= 2x^3 - 3x + 6 \quad \checkmark
 \end{aligned}$$

must have $\pm C$

$$\begin{aligned}
 \text{d) } \int_2^k (x-1) dx &= 4 \\
 \left[\frac{x^2}{2} - x\right]_2^k &= 4 \quad \checkmark \\
 \left(\frac{k^2}{2} - k\right) - \left(\frac{4}{2} - 2\right) &= 4 \quad \checkmark \\
 k^2 - 2k - 8 &= 0 \quad \checkmark \\
 (k-4)(k+2) &= 0 \\
 k &= 4, k = -2 \\
 \text{but } k &> 2 \\
 \text{So } k &= 4 \quad \checkmark
 \end{aligned}$$

14. a) i) $y = 1 - e^{-x}$

$$\frac{dy}{dx} = e^{-x}$$

when $x=1$, $\frac{dy}{dx} = \frac{1}{e}$ ✓

tangent has grad $\frac{1}{e}$, passes through $(1, 1 - \frac{1}{e})$ ✓

$$y - y_1 = m(x - x_1)$$

$$y - (1 - \frac{1}{e}) = \frac{1}{e}(x - 1)$$
 ✓

$$y - 1 + \frac{1}{e} = \frac{1}{e}x - \frac{1}{e}$$

$$y = \frac{1}{e}x + 1 - \frac{2}{e} \quad \text{or} \quad x - ey + e - 2 = 0$$

b) i) $y^2 + 4y - 3x - 5 = 0$

$$y^2 + 4y + 4 = 3x + 5 + 4$$

$$(y+2)^2 = 3(x+3)$$

$$(y+2)^2 = 4 \times \frac{3}{4}(x+3)$$

✓ for completing the square

✓ either of last two lines ok

ii) focal length: $\frac{3}{4}$

focus: $(-2\frac{1}{4}, -2)$ ✓

directrix: $x = -3\frac{3}{4}$ ✓

c) $y = \frac{2x^2}{e^{3x}}$

$$\frac{dy}{dx} = \frac{e^{3x} \cdot 4x - 2x^2 \cdot 3e^{3x}}{(e^{3x})^2}$$
 ✓

$$= \frac{4x - 6x^2}{e^{3x}}$$
 ✓

$$= \frac{2x(2-3x)}{e^{3x}}$$

OK to use product rule.

or $(4x - 6x^2)e^{-3x}$

$$d) \quad i) \quad \text{let } -x^2 + 4x - 1 = x^2 - 6x + 7$$

$$2x^2 - 10x + 8 = 0$$

$$x^2 - 5x + 4 = 0$$

$$(x-4)(x-1) = 0$$

$$x = 1, x = 4$$

$$y = 2, y = -1$$

So points of intersection are $(1, 2)$ and $(4, -1)$ ✓

$$ii) \quad \text{Area} = \int_1^4 ((-x^2 + 4x - 1) - (x^2 - 6x + 7)) dx$$

$$= \int_1^4 (-2x^2 + 10x - 8) dx \quad \checkmark$$

$$= \left[-\frac{2x^3}{3} + 5x^2 - 8x \right]_1^4$$

$$= \left(-\frac{128}{3} + 80 - 32 \right) - \left(-\frac{2}{3} + 5 - 8 \right)$$

$$= 9 \quad \checkmark$$

$$\text{So Area} = 9 \text{ units}^2$$

$$e) \quad i) \quad \frac{d}{du}(e^{x^2}) = 2xe^{x^2} \quad \checkmark$$

$$ii) \quad \int_0^1 6xe^{x^2} du = 3 \int_0^1 2xe^{x^2} du$$

$$= 3[e^{x^2}]_0^1 \quad \checkmark$$

$$= 3(e - 1) \quad \checkmark$$

15 a) i) $y = e^{-3x}$
 $y' = -3e^{-3x}$ ✓
 $y'' = 9e^{-3x}$ ✓

ii) Show $3y = 5y' + 2y''$

$$\text{LHS} = 3 \times e^{-3x}$$

$$= 3e^{-3x}$$

$$\text{RHS} = 5 \times (-3e^{-3x}) + 2 \times 9e^{-3x}$$

$$= 3e^{-3x}$$

So $\text{LHS} = \text{RHS}$

So $3y = 5y' + 2y''$ as req'd. ✓

b) $P(x, y)$, $A(8, -16)$

i) $PA = 3P_0$

$$PA^2 = 9P_0^2$$

$$(x-8)^2 + (y+16)^2 = 9[x^2 + y^2] \quad \checkmark$$

$$x^2 - 16x + 64 + y^2 + 32y + 256 = 9x^2 + 9y^2$$

$$8x^2 + 16x + 8y^2 - 32y = 320$$

$$x^2 + 2x + y^2 - 4y = 40 \quad \checkmark$$

$$x^2 + 2x + 1 + y^2 - 4y + 4 = 45$$

$$(x+1)^2 + (y-2)^2 = (\sqrt{45})^2$$

ii) So centre is $(-1, 2)$ radius $3\sqrt{5}$ units ✓ ($\sqrt{45}$ ok)

c) i) By Pythagoras, hypotenuse of triangular cross section is $5x$

$$\text{So } S = 2 \times \left(\frac{1}{2} \times 3x \times 4x \right) + (3x + 4x + 5x) \times h$$

$$= 12x^2 + 12xh \quad \text{as req'd.} \quad \checkmark$$

$$\text{ii) } V = \frac{1}{2} \times 3x \times 4x \times h \\ = 6x^2h$$

$$\text{from i) } 240 = 12x^2 + 12xh$$

$$20 = x^2 + xh$$

$$h = \frac{20 - x^2}{x} \quad \checkmark$$

$$\text{So } V = 6x^2 \times \frac{20 - x^2}{x}$$

$$= 6x(20 - x^2)$$

$$= 120x - 6x^3 \quad \text{as required } \checkmark$$

$$\text{iii) } \frac{dv}{dx} = 120 - 18x^2$$

$$\text{let } \frac{dv}{dx} = 0$$

$$120 - 18x^2 = 0$$

$$x^2 = \frac{20}{3}$$

$$x = \pm \frac{2\sqrt{5}}{\sqrt{3}} \quad \text{but } x > 0 \quad \checkmark$$

$$\frac{d^2v}{dx^2} = -36x$$

$$\text{when } x = \frac{2\sqrt{5}}{\sqrt{3}}, \quad \frac{d^2v}{dx^2} < 0$$

$$\text{So maximum occurs when } x = \frac{2\sqrt{5}}{3}$$

-1 for
not testing
nature

$$V = 120 \times \frac{2\sqrt{5}}{\sqrt{3}} - 6x \left(\frac{2\sqrt{5}}{\sqrt{3}} \right)^3$$

$$= \frac{240\sqrt{5}}{\sqrt{3}} - \frac{240\sqrt{5}}{3\sqrt{3}}$$

$$= \frac{160\sqrt{5}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{160\sqrt{5}}{3}$$

$$\text{So maximum volume is } \frac{160\sqrt{5}}{3} \text{ cm}^3. \quad \checkmark$$

$$d) i) V = \pi \int_0^6 x^2 dy$$

y	0	2	4	6
x	1	2	4	8
x ²	1	4	16	64

✓

$$\int_0^6 x^2 dy = \frac{2-0}{2} \{ 1 + 2 \times 4 + 2 \times 16 + 64 \} \quad \checkmark$$

$$= 105$$

✓

$$\text{So Volume} = 105\pi \text{ units}^3$$

ii) The estimation in part i) is an over-estimation as from the perspective of the y-axis, the curve is closer to the y axis than the straight line segments formed by joining the points. This relationship is preserved when the x values are squared in order to estimate the volume integral as the x values are larger than or equal to one.

✓ or similar.